Wing and Tail De-icing Boots (S.O.O. 6004 and Mod 6/2042) – Maintenance Practices

1. Removal/Installation

Equipment required:

- Toluol Solvent (Specification TT–T–548A)
- Pressure Squirt Gun
- Methyl-Ethyl-Ketone
- 1.1.1–Trichloroethane
- Sealant PR1422 – A2/B2 (PRC Corporation of Canada) or
- EC1675–A2/B2 (Minnesota Mining and Manufacturing Co.) or
- Pro-Seal 735 (Coast Pro-Seal Manufacturing Co. Calif)
- Adhesive Cement EC 1300L (Minnesota Mining and Manufacturing Co.)
- Adhesive Cement Bostik 1142 (B.B. Chemical Co.)
- One-inch Masking Tape
- 180-grit Emery Paper
- Trimming Tool SD5197 (Viking Air Limited Aircraft)
- Two-inch Rubber Roller
- Stitcher Roller
- Spatula

A. Remove Wing or Horizontal Stabilizer De-icer Boots

WARNING ADEQUATE VENTILATION MUST BE PROVIDED AND THE USE OF PROTECTIVE CLOTHING STRICTLY ADHERED TO AT ALL TIMES DUE TO THE TOXIC NATURE OF THE SOLVENTS.

Refer to Figure 1 for details.

Note To facilitate installation of boot, accurately mark position of boot centerline on adjacent wing or horizontal stabilizer at each end of boot. Also, accurately locate position of stall bars on wings, span and chordwise, by measuring to fixed datum points.

1) Loosen clamps and disconnect relevant de-icer hoses. Blank open ends and temporarily stow hoses.

2) Remove bolts securing wing fences, and release bond between wing fence seals and de-icer boots by continuous application of toluol at seam line when removing fences.

3) If de-icer boot is damaged, remove boot by peeling off in sections parallel to the inflatable tubes. If de-icer boot is apparently serviceable, remove boot as follows:

   a) Peel back one corner of upper trailing edge and apply toluol (toluene) while tension is being applied to peel back boot.

   b) Allow toluol to form a reservoir at seam line, then apply even tension to strip boot over leading edge at centerline. When removing bottom half of boot, use a pressure squirt gun to keep a supply of toluol at seam line.

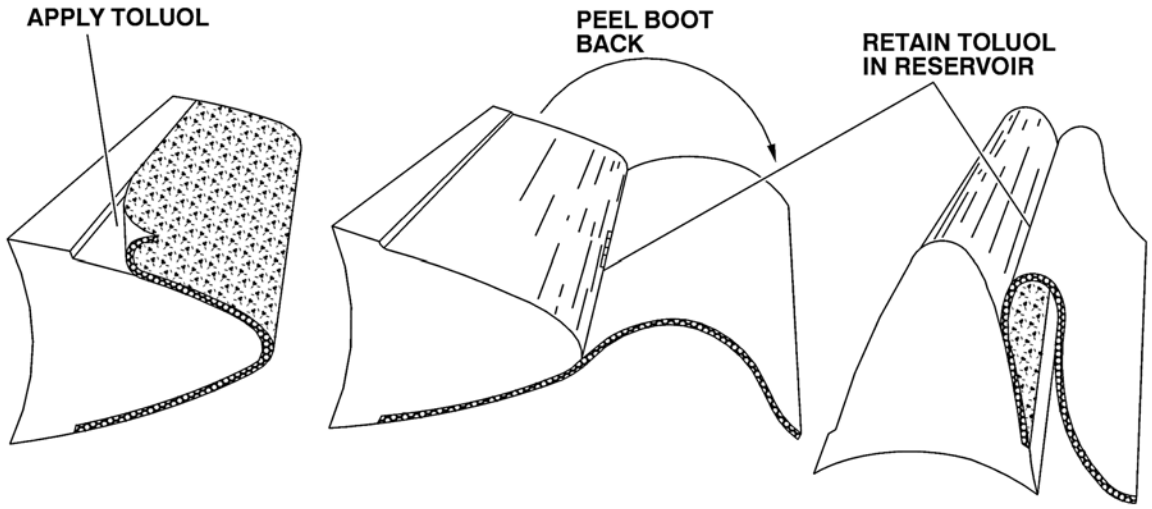
   c) Inspect inside of de-icer boot to ensure that boot was not damaged during removal.

   d) Thoroughly clean inside of boot using methyl-ethyl-ketone.

4) After removal of de-icer boot, clean leading edge as follows:

   a) Saturate a clean white cotton cloth in uncontaminated methyl-ethyl-ketone and thoroughly wash a small area at a time, wiping clean with a second, dry, clean cloth before solvent evaporates.

b) Keep cloth saturated and turn frequently to avoid using a dirty position. Pour solvent onto cloth to avoid contamination of solvent.



Removal of De-icer Boots
Figure 1

B. Install Wing or Horizontal Stabilizer De-icer Boots

WARNING **ADEQUATE VENTILATION MUST BE PROVIDED AND THE USE OF PROTECTIVE CLOTHING STRICTLY ADHERED TO AT ALL TIMES DUE TO TOXIC NATURE OF THE SOLVENTS.**

Refer to Figure 2 for details.

1) Thoroughly clean leading edge using methyl-ethyl-ketone, or toluol. Wipe dry with clean lint free cloth. Clean and wipe small areas at a time. Bond areas must be cleaned at least twice and must not be touched by bare hands or subjected to any other form of contamination.

2) Lay de-icer boot on clean flat surface and clean inside thoroughly with methyl-ethyl-ketone or toluol. Ensure that solvent does not come in contact with outer surface of boot.

3) Thoroughly stir adhesive EC 1300L.

4) Immediately after cleaning, apply a heavy coat of adhesive to leading edge (except anchor nut holes on horizontal stabilizer), and to fabric backing of de-icer boot. Apply adhesive as evenly as possible without excessive brushing. Care must be taken to cover all areas sufficiently.

CAUTION **DO NOT APPLY ADHESIVE WHEN RELATIVE HUMIDITY EXCEEDS 90 PERCENT.**

5) Allow adhesive to dry for one hour (below 75 percent relative humidity) or two hours (75 to 90 percent).

6) Using a non-greasy marker, draw line on leading edge between reference marks made during removal, refer to Para 1.A.

7) Similarly, re-mark along centerline of inside of de-icer boot.

8) Apply a second coat of adhesive to both de-icer boot and leading edge.

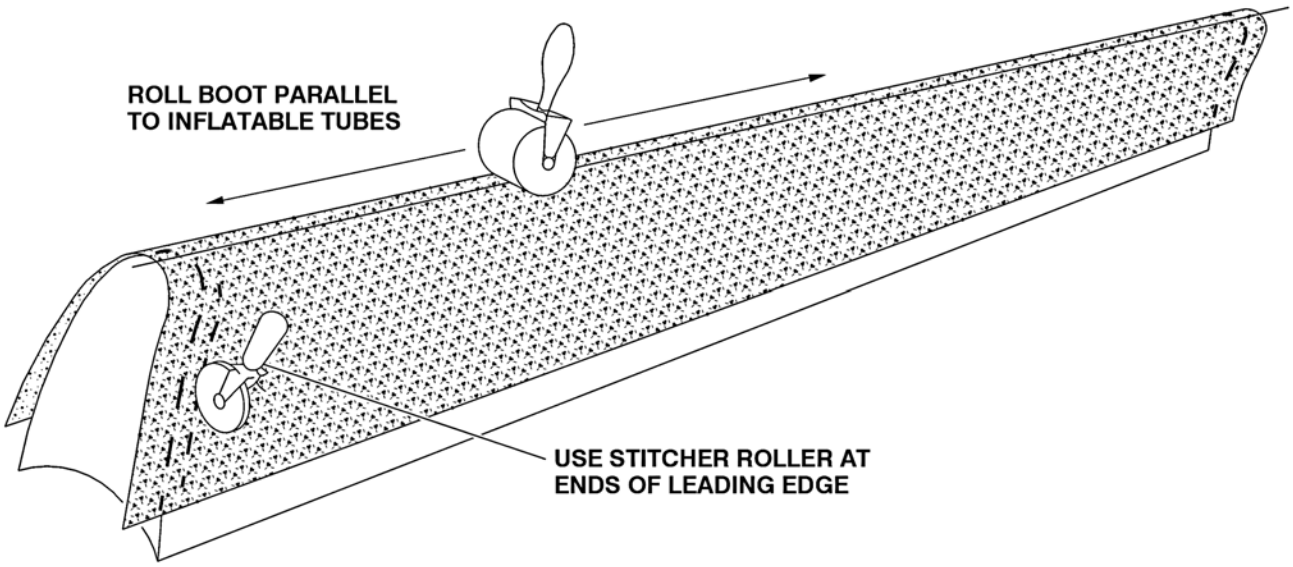
9) Allow second coat of adhesive to dry for a minimum period of two hours.

   Note Up to 48 hours drying time is permissible for each coat of adhesive provided the surface is protected against contamination by covering it with kraft paper. Avoid touching adhesive coating.

10) Position de-icer boot on leading edge, utilizing marks made during removal to position boot. Ensure air connector or connectors are centrally located in skin hole. Reactivate adhesive in area around air connectors and bond. Reactivate adhesive by damping de-icer boot and leading edge using a clean lint-free cloth moistened with toluol. Avoid rubbing, as this could remove adhesive.

11) Progressively unroll de-icer boot reactivating and bonding a small area at a time along centerline, matching centerline of de-icer boot with line on leading edge; press down firmly by hand to remove all air bubbles. Improve bond by rolling out spanwise using a 2–inch rubber roller. Do not bond within two inches of the edge until remainder of de-icer boot has been bonded and all air pockets have been removed. Do not bond ends of boots until any necessary trimming has been done.

    Note Entrapped air may be removed by using a thin spatula and sliding it between bonded surfaces into the air bubble, press out and remove spatula simultaneously. Where this is impractical, release boot (using toluol as necessary) as far as the affected area and re-bond boot.



Use of Rubber Rollers to Improve Bond
Figure 2

12) Horizontal stabilizer de-icer boot trimming:

    a) Trim inboard end of de-icer boots to conform with leading edge structure assembly, using tool SD5197 or a sharp knife, and trim outboard end so that boot overlaps onto tip by approximately 1/2–inch.

    b) Pierce de-icer boot in middle of anchor nut holes and drill out boot using a 3/16–inch diameter drill. Do not damage anchor nut threads.

    c) On lower inboard end of boots, use tool SD5197 or a sharp knife to trim edge to accommodate installation of vortex generators.

13) Wing de-icer boot trimming:

a) Trim inboard end of wing boot to landing lamp and outboard end to wing tip joint.

b) On left-hand wing boot, remove with tool SD5197 or a sharp knife, the portion of the boot covering cut out for lift detector installation. Cut out boot so as to leave a gap of approximately 1/32–inch between boot and lift detector mounting plate.

14) Ensure bond is intact in areas that have been trimmed or cut out.

15) Seal gap between left-hand wing de-icer boot and lift detector mounting plate as follows:

a) Clean area with 1,1,1–trichloroethane soaked cloth and dry with clean cloth before solvent evaporates. Pour solvent into cloth so as to maintain clean solvent supply. Protect cleaned area against recontamination and do not touch with bare hands.

b) Apply sealant PR1422–B2 or equivalent. Press sealant into gap and remove surplus with spatula. Fair sealant at joint edges with rubber tipped smoothing tool (refer 20-40-21, Sealants - Weather Sealing – General Data).

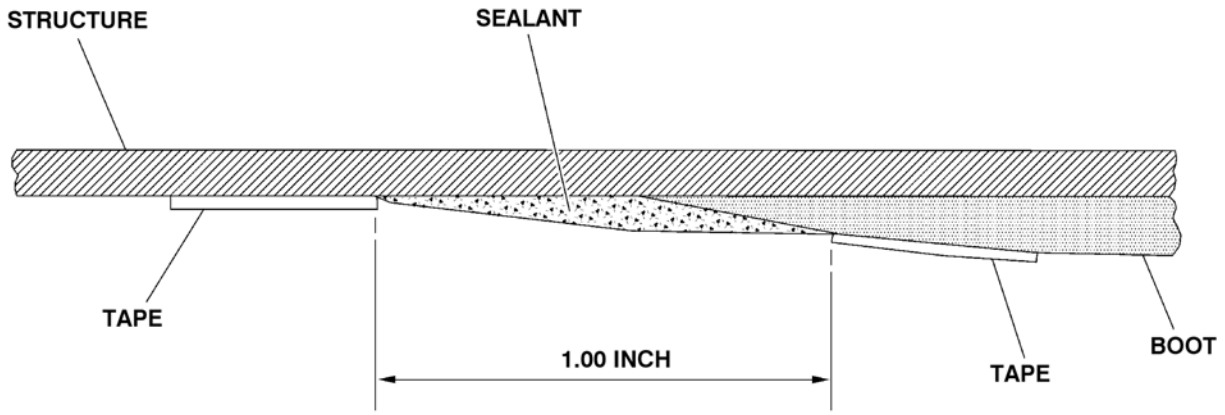
c) Remove any sealant smears with cloth moistened with methyl-ethyl-ketone. Ensure that solvent does not penetrate between cemented edges of boot.

16) Allow boot to cure for minimum of 24 hours at room temperature prior to following operations.

Note Following an initial cure of 1 hour at room temperature, curing time can be reduced by application of heat to de-icer boot area for 2 hours at 110 to 115°F (43 to 46°C). Heat source may be 250W heat lamps placed 10 to 12 inches (25 to 30 cm) from inner surface of leading edge. To equalize heat distribution, aluminum foil should be wrapped around de-icer boot. Curing time of sealants will also be reduced by application of heat.

17) Inflate boot to 15 psi with nitrogen or clean, dry air and check for proper inflation, adhesion and leaks. Leak rate must not exceed one psi in 60 seconds.

18) Refer to Figure 3. Apply masking tape to boot and structure as shown and feather seal thin bead of sealant PR1422–B2 or equivalent along boot edge using suitable spatula. Remove masking tape on completion of sealing operation.



Typical Feathered Edge Seal
Figure 3

19) Install stall bar to wings as follows:

    a) Vigorously scrub boot with methyl-ethyl-ketone in area where stall bar is to be bonded.

    b) Allow to dry.

    c) Locate and mark stall bar position as noted during removal, refer to Para 1.A.

    d) Apply two coats of EC 1300L adhesive to both stall bar and the surface of boot to which bar is to be bonded as in Para 1.B.3) through Para 1.B.5), Para 1.B.8) and Para 1.B.9).

    e) Reactivate cement of bar and boot using clean lint-free cloth moistened with toluol. Avoid rubbing, as this could remove adhesive.

    f) Firmly press stall bar on to boot and use a rubber roller on leading edge of bar to seat it.

    g) Clean up any excess adhesive around edge of stall bar with a toluol-soaked rag. Daub the area gently and take care not to allow any toluol to run down between boot and stall bar.

20) Remove blanks from de-icer boot air connections and end of hoses. Connect hoses and secure using clamps.

21) Operate de-icer system and check de-icer boots for bubbles.

22) Install wing fences as follows:

    a) Clean fence seal and area of boot contacted by seal with a toluol-soaked cloth, and wipe dry with a clean cloth before solvent evaporates.

    b) Apply by brush, a thin, even coat of Bostik 1142 to both bonding surfaces, making as few strokes as possible. Stir adhesive thoroughly immediately before use.

    c) Allow adhesive to dry until tacky.

    d) Position wing fence on wing and insert attachment bolts. Press seal in contact with boot using fingers or rubber roller and tighten attachment bolts.

23) Functionally test wing and tail de-icing system (refer to 30-10-00, Wing and Tail De-icing System (Mod S.O.O. 6187 and Mod 6/2042) – General Data).

24) Verify a conductive path exists between the de-ice boot, the conductive edge sealer and the skin. Using a Resistivity Meter, measure the conductivity, acceptable range is 10^3 to 10^5 ohms.

    Note Resistivity Meter (Monroe Electronic Model 291 or equivalent) contains two parallel bars on the underside, one is placed on the boot and the other is placed on the edge sealer.

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